

Paul Prae

Principal AI Engineer & Architect | Buford, GA | hireme@paulprae.com | linkedin.com/in/paulprae | github.com/praeducer | paulprae.com

PROFESSIONAL SUMMARY

AI and data engineering leader with 13+ years building enterprise solutions across healthcare, life sciences, and financial services. Currently managing Arine's enterprise data platform on Snowflake and AWS, supporting a medication-optimization platform serving 45+ health plans and >30 million members. Previous roles span enterprise AI/ML solutions architecture at Amazon Web Services, healthcare AI practice leadership at Booz Allen Hamilton (\$9B+ revenue), and automated ML pipeline development at NeuroLex Labs. Brings deep expertise in AI agents, cloud-native data platforms, and HIPAA-compliant systems alongside engineering leadership across data operations, platform architecture, and team development.

PROFESSIONAL EXPERIENCE

Staff AI DataOps Engineer

Arine | Remote | Sep 2025 – Present

- Managed enterprise data platform on Snowflake and AWS, processing petabytes of healthcare data from hundreds of sources for a medication-optimization platform serving 45+ health plans and >30M members (Inc. 5000 #5 in AI)
- Designed the AI DataOps Toolkit — an agentic framework with HIPAA-compliant coding assistants including a Data Engineering Agent for autonomous ETL pipeline generation and a Dev Environment Agent that eliminated multi-day developer onboarding
- Built observability pipeline monitoring Arine's autonomous pharmacist AI agent, ingesting production behavior data from DynamoDB into Snowflake, transforming with dbt, and surfacing metrics in QuickSight dashboards
- Led AI enablement workshops training groups of up to 100 engineers on AI-first development practices and tools
- Engineered high-throughput CDC pipelines from PostgreSQL to Snowflake using serverless, event-driven architectures (Lambda, Step Functions, EventBridge)

Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, GA | Jul 2024 – Mar 2025

- Led AI and analytics solution development at a Fortune 500 consulting firm (27,000+ employees, \$9B+ revenue), designing products leveraging clinical, behavioral, and genomic data for population health and predictive medicine
- Delivered AI products across FedRAMP and HIPAA-compliant environments, mitigating algorithmic bias and privacy risks for healthcare and life science clients
- Drove cross-functional collaboration with engineers, data scientists, and domain experts to address clinical, regulatory, and technological requirements

Principal AI Architect, Founder

Modular Earth | Buford, GA | Dec 2022 – Present

- Launched a not-for-profit building open-source AI agents for working-class families, delivering career planning and financial advisory tools using Ollama, LangChain, n8n, and Neo4j
- Designed local-first, privacy-preserving multi-agent architectures using Tauri, Rust, and TypeScript that operate without cloud infrastructure dependencies
- Architected a 6-stage agent workflow with role-based permissions, structured handoffs, and knowledge management for decentralized AI applications

Neuroinformatics Data Platform Architect

TReNDS Center | Atlanta, GA | Jan 2022 – Sep 2023

- Architected AWS cloud infrastructure for COINSTAC, an open-source platform enabling federated neuroimaging and genomics research across institutions with differential privacy protections for PHI
- Built GPU compute infrastructure for deep learning workloads and drove commercialization through SBIR grant writing, open-source contributor recruitment, and industry partnerships
- Spearheaded product roadmap as Scrum master, contributing to peer-reviewed research published in Frontiers in Neuroinformatics (2023)

Chief AI Officer, Founder

Hyperbloom | United States | Jan 2020 – Sep 2025

- Grew AI consulting business to \$1.4M ARR from a \$40K investment over 5+ years, delivering cloud-based AI and data solutions across healthcare, life science, and financial services
- Architected disaster recovery solutions for clinical trial software supporting 10,000+ research sites across 45 countries with 5.5M monthly remote monitoring activities
- Designed a blockchain-based data platform for secure life science data exchange, meeting CCPA, GDPR, and HIPAA requirements with HL7 and CDISC interoperability standards

Enterprise AI and ML Solutions Architect

Amazon Web Services | Greater Atlanta Area | Aug 2018 – May 2021

- Managed a portfolio of 10+ enterprise accounts including Cox Enterprises, Cox Automotive, Equifax, NCR, Accenture, and Deloitte, delivering AI/ML architecture guidance and driving AWS adoption
- Created technical white papers, reference architectures, and presentations; engaged nationally as a subject matter expert in AI and machine learning
- Built deep relationships with C-suite executives to define technical and business strategies encompassing go-to-market planning, cost management, and technical recruiting

Senior AI Engineer

NeuroLex Labs | Greater Atlanta Area | Jan 2018 – May 2020

- Architected automated ML pipelines for healthcare voice computing — collecting, cleaning, training, and deploying models processing text and audio data for neurodegenerative disease prediction
- Drove TDD adoption across the engineering team with 4 testing types (regression, functional, load, integration); mentored international fellows in the Tribe program for neurodegenerative disease research

Advanced Analytics Consultant

Slalom Consulting | Greater Atlanta Area | Jul 2015 – Jan 2018

- Delivered 7+ client engagements generating \$2M+ in consulting revenue, including a statewide behavioral health demand forecasting system across dozens of facilities in Georgia
- Deployed data science experiments into cloud-based production systems on Microsoft Azure; recognized nationally as a subject matter expert in NLP and machine learning

Software Engineer

Red Ventures | Fort Mill, SC | Mar 2014 – Jul 2015

- Engineered full-stack marketing platform on the LAMP stack serving 16+ clients including DirecTV, Verizon, AT&T, Liberty Mutual, and Progressive, developing proprietary CMS, BI framework, and hundreds of high-traffic websites

Senior Support Engineer

Microsoft | Charlotte, NC | Jul 2012 – Mar 2014

- Automated deployment and testing workflows using PowerShell and C# while delivering advanced technical support to Fortune 100 customers including Chevron, Walmart, and Intel; nominated for “Entrepreneur of the Year” (2013)

EDUCATION

Bachelor of Science, Computer Science (AI Emphasis) & Bachelor of

Arts, Cognitive Science (AI Foundation) **University of Georgia** | 2008 – 2012 Graduated Cum Laude. Dean’s List. HOPE Scholarship.

TECHNICAL SKILLS

AI & Machine Learning: Large Language Models, Multi-agent Systems, Generative AI, Natural Language Processing, MLOps, Predictive Analytics, Prompt Engineering, Conversational AI, Deep Learning

Cloud & Infrastructure: AWS (Bedrock, SageMaker, ECS/EKS, Lambda, Step Functions, S3, DynamoDB, QuickSight), Azure (Machine Learning, Bot Service, Cognitive Services), Snowflake, Docker

Data Engineering: dbt, Apache Kafka, PostgreSQL, SQL Server, MySQL, Neo4j, ETL/CDC Pipelines, Data Governance, HIPAA/GDPR Compliance

Programming Languages: Python, SQL, JavaScript, C#, Bash, PowerShell, Rust, TypeScript

AI Tools & Frameworks: Cursor AI, Claude Code, GitHub Copilot, LangChain, Ollama, n8n, Open WebUI, MCP Tools

Leadership & Strategy: Engineering Team Management, AI Strategy & Governance, Technical Coaching, Solutions Architecture, Pre-Sales Engineering, Agile/Scrum

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
- **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)
- **AWS Certified Machine Learning - Specialty** — Amazon Web Services (Jan 2020)
- **AWS Certified Solutions Architect** — Amazon Web Services (Jul 2018)

PUBLICATIONS

[Enhancing Collaborative Neuroimaging Research: COINSTAC Vaults for Federated Analysis](#) *Frontiers in Neuroinformatics* (Jun 2023). Co-authored research on federated analysis for collaborative neuroimaging, enabling researchers to analyze datasets across institutions without sharing raw data using differential privacy.